TECHNICAL SUPPLEMENT 04 · ENGLISH

Mission Outcomes and Human-Proximity Events

Agri-Social Navigation in Vineyard Corridors

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Study identity	Specification
Campaign / Campaña	EVAL_20261019_112257
Navigation / Navegación	9 × 4 × 10 = 360; SUCCESS 266; TIMEOUT 94
Benchmark	180; 20 per method / por método
Version / Versión	20 October / octubre 2026

Numerical companion to the revised Sensors manuscript. This report uses the identified endpoint dataset and separates it from the older public campaigns. All seven authors are listed in the original manuscript order.

1. Planned denominators retained

Method	n	SUCCESS	TIMEOUT	Success rate
M0	40	40	0	100.0%
M1	40	0	40	0.0%
M2	40	0	40	0.0%
M3	40	36	4	90.0%
M4	40	40	0	100.0%
B1	40	35	5	87.5%
B2	40	39	1	97.5%
B3	40	40	0	100.0%
B4	40	36	4	90.0%

TIMEOUT is a recorded terminal mission failure, not a pending or missing run. It remains in the 360-attempt denominator and is never replaced by a successful repeat. A proximity event is not a collision or a validated comfort measure.

Failure - exact count intervals / intervalos exactos

E	Method	Events / n	Risk	95% exact CI
E1	M0	0/10	0.000	[0.000, 0.308]
E2	M0	0/10	0.000	[0.000, 0.308]
E3	M0	0/10	0.000	[0.000, 0.308]
E4	M0	0/10	0.000	[0.000, 0.308]
E1	M1	10/10	1.000	[0.692, 1.000]
E2	M1	10/10	1.000	[0.692, 1.000]
E3	M1	10/10	1.000	[0.692, 1.000]
E4	M1	10/10	1.000	[0.692, 1.000]
E1	M2	10/10	1.000	[0.692, 1.000]
E2	M2	10/10	1.000	[0.692, 1.000]
E3	M2	10/10	1.000	[0.692, 1.000]
E4	M2	10/10	1.000	[0.692, 1.000]
E1	M3	0/10	0.000	[0.000, 0.308]
E2	M3	0/10	0.000	[0.000, 0.308]
E3	M3	0/10	0.000	[0.000, 0.308]
E4	M3	4/10	0.400	[0.122, 0.738]
E1	M4	0/10	0.000	[0.000, 0.308]
E2	M4	0/10	0.000	[0.000, 0.308]
E3	M4	0/10	0.000	[0.000, 0.308]
E4	M4	0/10	0.000	[0.000, 0.308]
E1	B1	0/10	0.000	[0.000, 0.308]
E2	B1	0/10	0.000	[0.000, 0.308]
E3	B1	0/10	0.000	[0.000, 0.308]

E	Method	Events / n	Risk	95% exact CI
E4	B1	5/10	0.500	[0.187, 0.813]
E1	B2	0/10	0.000	[0.000, 0.308]
E2	B2	0/10	0.000	[0.000, 0.308]
E3	B2	0/10	0.000	[0.000, 0.308]
E4	B2	1/10	0.100	[0.003, 0.445]
E1	B3	0/10	0.000	[0.000, 0.308]
E2	B3	0/10	0.000	[0.000, 0.308]
E3	B3	0/10	0.000	[0.000, 0.308]
E4	B3	0/10	0.000	[0.000, 0.308]
E1	B4	0/10	0.000	[0.000, 0.308]
E2	B4	0/10	0.000	[0.000, 0.308]
E3	B4	0/10	0.000	[0.000, 0.308]
E4	B4	4/10	0.400	[0.122, 0.738]

IntimateEvent - exact count intervals / intervalos exactos

E	Method	Events / n	Risk	95% exact CI
E1	M0	0/10	0.000	[0.000, 0.308]
E2	M0	0/10	0.000	[0.000, 0.308]
E3	M0	0/10	0.000	[0.000, 0.308]
E4	M0	0/10	0.000	[0.000, 0.308]
E1	M1	0/10	0.000	[0.000, 0.308]
E2	M1	0/10	0.000	[0.000, 0.308]
E3	M1	0/10	0.000	[0.000, 0.308]
E4	M1	0/10	0.000	[0.000, 0.308]
E1	M2	0/10	0.000	[0.000, 0.308]
E2	M2	0/10	0.000	[0.000, 0.308]
E3	M2	0/10	0.000	[0.000, 0.308]
E4	M2	0/10	0.000	[0.000, 0.308]
E1	M3	0/10	0.000	[0.000, 0.308]
E2	M3	0/10	0.000	[0.000, 0.308]
E3	M3	0/10	0.000	[0.000, 0.308]
E4	M3	0/10	0.000	[0.000, 0.308]
E1	M4	0/10	0.000	[0.000, 0.308]
E2	M4	0/10	0.000	[0.000, 0.308]
E3	M4	0/10	0.000	[0.000, 0.308]
E4	M4	0/10	0.000	[0.000, 0.308]
E1	B1	0/10	0.000	[0.000, 0.308]
E2	B1	0/10	0.000	[0.000, 0.308]
E3	B1	0/10	0.000	[0.000, 0.308]

E	Method	Events / n	Risk	95% exact CI
E4	B1	0/10	0.000	[0.000, 0.308]
E1	B2	0/10	0.000	[0.000, 0.308]
E2	B2	0/10	0.000	[0.000, 0.308]
E3	B2	0/10	0.000	[0.000, 0.308]
E4	B2	0/10	0.000	[0.000, 0.308]
E1	B3	0/10	0.000	[0.000, 0.308]
E2	B3	0/10	0.000	[0.000, 0.308]
E3	B3	0/10	0.000	[0.000, 0.308]
E4	B3	0/10	0.000	[0.000, 0.308]
E1	B4	0/10	0.000	[0.000, 0.308]
E2	B4	0/10	0.000	[0.000, 0.308]
E3	B4	0/10	0.000	[0.000, 0.308]
E4	B4	0/10	0.000	[0.000, 0.308]

Clopper-Pearson intervals describe each observed binomial count under the working independent-trial assumption. With zero events in ten runs, the exact two-sided 95% upper limit is 0.3085. Zero observed events therefore do not establish equivalence or zero risk.

Source and scope

Source: data/; results/expected/. PDFs provide the reading reports; CSV/JSON retain numerical inputs and provenance. Scope: evaluated simulation, without new experiments, independent raw-stream reconstruction or physical validation.